

AI Project Ideas - LangChain & RAG

1. Chat with PDF (Document QA)

Objective: Create an AI chatbot that can read and answer questions from PDF documents.

Tech Stack: LangChain, OpenAI API (GPT), PyPDF2, or other PDF parsing libraries.

Description: Extract text from a PDF document and allow users to ask questions. The model retrieves relevant sections of the document and answers queries based on the content.

2. Personalized AI Tutor for Textbooks

Objective: Build an AI tutor that can teach concepts from a specific textbook or course material.

Tech Stack: LangChain, OpenAI API (GPT), Knowledge Base, Retrieval-Augmented Generation.

Description: Integrate textbooks, course notes, or research papers, and let students interact with an AI to understand complex topics. The AI answers based on the content in the material.

3. Multilingual Translation and Summarization Assistant

Objective: Create an AI assistant that can translate and summarize documents in multiple languages.

Tech Stack: LangChain, OpenAI API, multilingual datasets.

Description: Users can upload documents in different languages and the assistant will provide translations and concise summaries while maintaining the meaning of the original text.

4. AI Content Creation Assistant

Objective: Build an AI that helps creators (bloggers, journalists, writers) create content by suggesting articles, summaries, and generating ideas.

Tech Stack: LangChain, OpenAI API, content API.

Description: This tool can pull articles from various sources, summarize them, or even suggest headlines and ideas based on a given topic or document.

5. Chat with MDN Docs (Web Development Assistant)

Objective: Create a chatbot that can query and interact with MDN (Mozilla Developer Network) documentation to assist developers.

Tech Stack: LangChain, OpenAI API, MDN API or Web Scraping.

Description: Developers can chat with the AI assistant to quickly get information, examples, and explanations about web development concepts from the MDN documentation.